QUALYXUS

Air-Gapped Edge AI for Visual Inspection & Process-Quality Intelligence

- **Founder & Lead Architect:** Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

> **We solve** late-stage scrap and escaped defects **by providing** an on-premise edge-AI platform that inspects every part and links each defect to the machine cycle that produced it, **to help** automotive, aerospace & defense, and industrial manufacturers **trace** defects to process drift and machine wear, so causes get fixed sooner, with all data staying inside the plant.

The Problem

Quality and Machine Data Live in Separate Silos

Plants run two disconnected systems, so the defect and its cause are rarely seen together:

1. Machine Vision (Post-Mortem Sorting)

- Inspects parts **after** they are made, so scrap is caught only after material, machine hours and labour are spent.
- Rule-based setups need re-programming for every new part and struggle with variable cosmetic defects.
- Says pass or fail, but not **why**: blind to the machine conditions that caused the defect.

2. Machine Telemetry (Blind to Outcomes)

- Monitors temperatures, pressures, spindle load and vibration to prevent breakdowns.
- Never learns which parts came out bad, so it cannot tell which signal patterns cause defects.
- A machine can stay inside every alarm limit while slowly drifting into producing scrap.

Target Customer Profile: Plant Heads, VPs of Quality and Maintenance Heads running automated machining, moulding, casting, stamping, welding and assembly lines in automotive, aerospace & defense, and industrial manufacturing.

The Solution

Every Defect Linked to Its Machine Cycle

Qualyxus fuses camera inspection with machine-cycle data on one on-premise edge box, so quality and maintenance teams see the same cause.

Camera + Machine Data
Industrial cameras; Modbus TCP / OPC UA

On-Prem Edge Engine
Inspect, link to cycle, score quality & wear

Advisory PLC Signal
One verdict per part; plant logic decides

Core Deep-Tech Innovations

1 Learns from Good Parts

Anomaly detection (Dinomaly / PatchCore on DINOv2) trains on several hundred good parts, with no defect library needed. AUROC 0.95–0.99 on 4 categories of the public VisA benchmark.

2 Calibrated False Rejects

Reject thresholds are set by statistical (conformal) calibration with a stated confidence bound, per part across all cameras, so the plant knows its false-reject rate before go-live.

3 Root Cause & Machine Health

Each part is linked to the cycle that made it. A wear index per component tracks drift (heater bands, check ring, cooling), with approval-gated fixes that cut scrap 79% in simulation.

4 Air-Gapped by Design

Runs fully on-premise on an NVIDIA GPU, verified offline with no downloads. Shadow mode, one-command rollback and a tamper-evident audit log built in.

Built & Measured

Working Prototype, Pre-Revenue: What the Platform Already Does

0.95–0.99

Anomaly-detection AUROC on 4 categories of the public VisA benchmark (previous baseline: 0.56–0.96). With thresholds set for low false rejects, it caught 35–90% of defects, depending on the part.

~1%

False rejects per part, calibrated across a simulated six-camera station: 2 of 400 unseen good parts, with all 4 defect types caught (synthetic).

100% → 0%

False rejects on parts rotated 8° after automatic part registration, with all defects still caught; 15% at 15° (synthetic).

0.83

Mask IoU for defect segmentation (RF-DETR-Seg): 100% recall and defect-type accuracy, 16.5 ms per image on a laptop GPU (synthetic).

~4 s

To promote a new model live with no downtime; 98.5% shadow-mode agreement in closed-loop simulation.

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Automated tests passing, including forced failures: dead cameras, PLC outage, power cuts. End-to-end offline runs keep the audit log intact (simulation).

Results on the public VisA benchmark and on our synthetic part and process simulator. Nothing has run on a real PLC, camera, Jetson or factory line yet: measuring that is the goal of the paid pilots.

Traction

Early Industry Conversations

I have spoken with the heads at three manufacturers about piloting Qualyxus on their parts.

Rotary Connectors

- **Pilot:** spoke with the heads about testing Qualyxus on their parts.
- **360° inspector:** agreed to help build the design, starting as soon as funding arrives.

Rossell Techsys

- **Pilot:** spoke with the heads about testing Qualyxus on their parts.

Moog

- **Pilot:** spoke with the heads about testing Qualyxus on their parts.

Once funded: the 360° inspector design starts with Rotary Connectors and the test rig gets built for trials on real parts. **Next Milestone:** turn these conversations into 60-day paid pilots and publish measured results.

Conversations and design support so far. No signed pilots or revenue yet.

Competition

Where Qualyxus Fits

	Legacy Vision (Cognex / Keyence)	AI Vision Competitors (SwitchOn)	Heavy Automation (Siemens / Bosch)	QUALYXUS PLATFORM
Operational Method	Post-Mortem Sorter (catches scrap late)	Primarily Visual Inspection	Heavy Physical Sensor Wiring	**Inspection + Process Drift & Wear Tracking**
Data Underpinnings	Rule-Based Image Tools (+ add-on deep learning)	Deep-Learning Image Inspection	Machine Telemetry Logs	**Images Linked to Machine-Cycle Data**
Deployment Time	Weeks of Integrator Programming per Part	Under 45 Min per New SKU (Vendor Claim)	Multi-Month Projects	**Target: 1–2 Day Install (No Machine Rewiring)**
Safety Compliance	N/A (Post-Line)	N/A (Visual Check)	Changes May Need Safety Re-validation	**Advisory Only (No Safety-Function Changes)**
Budget Captured	Quality Control Only	Primarily Quality Control	Maintenance Only	**Dual Budget: Quality + Maintenance**
Typical Cost to Plant (indicative)	₹1.2L – ₹7.5L per smart camera + integration (one-time)	Subscription (pricing not public)	₹50L+ CapEx projects	**₹2–2.5L setup + ₹20k / month per line**

Cognex In-Sight 2000 to 8000-series dealer listings in India (ex-tax, lens and integration extra). Ranges are indicative. Others also link quality to process data (e.g. Kistler ComoNeo, Eigen, Sciemetric); our difference is learning from every inspected part, on-premise, at retrofit cost.

Market Opportunity

TAM / SAM / SOM — Bottom-Up Sizing

$45B

TAM (Total Addressable Market)

~₹3.9 lakh crore — Global AI visual inspection (~$31B) + predictive maintenance (~$14B), 2025 estimates.[1]

$590M

SAM (Serviceable Addressable Market)

~₹5,100 Cr / yr — ~1.7 lakh inspection points on machining, moulding, casting & forging, stamping, welding and assembly lines (~1.3 lakh India, ~0.4 lakh SE Asia) × ~₹3L blended contract.[2]

$5.5M

SOM (Serviceable Obtainable Market)

~₹48 Cr ARR — 2,000 lines in Indian industrial corridors (Pune, Chennai, Bengaluru, NCR, Hyderabad, Hubballi–Belagavi) × ₹2.4L list / yr. 5-year target (~1.5% of India); exports additive.

Market Tailwinds & Growth Drivers

Auto-Component Boom

₹7.6 lakh crore turnover in FY26, up 12.7% year on year, as global OEMs shift sourcing to India.[3]

India PLI Expansion

₹1.97 lakh crore PLI outlay across 14 sectors, incl. auto components, ACC batteries and electronics.[4]

Large Installed Base

₹31,781 Cr of machine tools consumed in FY25[5], ~5,000 foundries and 500+ forging units[6]: retrofit beats ₹50L+ replacement.

[1] MarketsandMarkets, Market Research Future; IMARC. [2] Qualyxus estimate. [3] ACMA, Jul 2026. [4] PIB, Nov 2024. [5] IMTMA (provisional). [6] Foundry Informatics Centre; AIFI. ₹87 = $1.

Business Model

Hybrid Hardware-Enabled SaaS

How we land: 60-day paid pilot beside manual inspectors (₹1L, credited on conversion), then 5–8 lines per plant.

1. One-Time Setup & Hardware: ₹2.0L – ₹2.5L per line

- Single-camera station: NVIDIA Jetson Orin-class edge computer, industrial camera + lens, lighting, enclosure, target 1–2 day install.
- Multi-camera "All-Faces" station: ₹5L – ₹7L.
- Target hardware gross margin: **20–30%**

2. Software Licence: ₹20,000 / month per line

- ₹2.4L / year, billed annually: models, retraining, updates, dashboard, on-site support.
- Costs no more than one inspector (₹20–35k / month loaded), yet covers all three shifts.
- Target software gross margin: **80%+**

Customer Payback (per line)

Metric	Value
Annual value delivered	₹6L – ₹15L typical; up to ₹25L on high-value parts
Year-1 cost to customer	₹4.4L – ₹4.9L (hardware + licence)
Payback period	4 – 10 months
Year-2+ return on licence	2.5× – 6×

Qualyxus Unit Economics (targets)

Metric	Target
Lines per plant (within 24 months)	5 – 8
Customer acquisition cost per plant	₹5L – ₹8L (pilot + field sales)
Gross profit per plant at 6 lines	~₹11.5L / year
CAC payback	5 – 9 months

All figures are planning assumptions to be validated in paid pilots.

Team & Advisors

Leadership, Cap Table & Strategic Advisors

Founder & Core Architect

Utkarsh Sengar (B.Tech CSE, IIT Dharwad)

- Built the full platform: good-parts-only anomaly detection, defect segmentation, part-to-cycle linkage and risk models, edge deployment and model operations.
- LinkedIn: linkedin.com/in/utkarsh-sengar

Cap Table Status

100% Founder Owned (Utkarsh Sengar)

Company incorporation planned with dhaRti support.

Strategic Advisory Board

Dr. Rakesh Kumar

VP R&D, Rossell Techsys (ex-Moog, ex-Indian Navy).

Shankar Lal

Discipline Chief, Cyient (21 years at HAL).

Ramesh Rana

Operations Lead, SIG SAUER.

How dhaRti Can Help

What I Need to Reach the First Paid Pilots

1 Incubation & Lab Space

A campus desk, a lab bench for a test inspection rig (camera, lighting, sample parts) and GPU compute time.

Unlocks: Real-hardware testing before plant pilots.

2 Seed Support: ₹5L to Start

Up to ₹15L in milestone-based tranches (next slide), from dhaRti directly or through NIDHI-PRAYAS / Startup India Seed Fund with dhaRti as host.

Unlocks: First 3 pilot stations and Jetson benchmarks.

3 Pilot Introductions

Introductions to 3–5 plants in the Hubballi–Dharwad–Belagavi cluster: foundries, machining shops and aerospace suppliers.

Unlocks: First paid 60-day pilots, close to campus.

4 Mentors

A manufacturing-quality expert, an industrial-sales mentor and an IP advisor.

Unlocks: Faster, better-aimed pilots and pricing.

5 IP & Company Setup

Patent assessment and filing for our part-to-cycle linking and calibration methods; incorporation and compliance guidance.

Unlocks: Defensible IP before wider rollout.

6 Investor Access

Introductions to angels and funds for the ₹2 Cr pre-seed once pilots are live.

Unlocks: The ₹2 Cr pre-seed after pilots show results.

In return: an IIT Dharwad deep-tech startup in manufacturing AI, with measured results from real plants within 9 months of incubation and monthly progress reports to dhaRti.

Seed Support Request

₹5 Lakh to Start; Up to ₹15 Lakh over 9 Months as Milestones Are Met

Use of Funds

Item	Amount
3 pilot inspection stations (camera, lighting, Jetson)	₹5.0L
Lab test rig + Jetson development kit	₹1.5L
Engineering intern / junior engineer, 6 months	₹3.0L
Plant installs and travel	₹1.5L
Incorporation, DPIIT recognition, provisional patent	₹1.5L
Compute, software and tools	₹1.0L
Contingency	₹1.5L
Total	₹15.0L

Released Only as Milestones Are Met

Tranche 1 · ₹5L · Month 0

Test rig, first station, 360° inspector design start, company setup and patent filing.
Milestone (month 3): rig live; Jetson speed and accuracy benchmarks published.

Tranche 2 · ₹5L · Month 3

Two more stations and the intern's first 3 months.
Milestone (month 6): 3 pilot stations installed in nearby plants.

Tranche 3 · ₹5L · Month 6

Intern months 4–6, plant installs and travel, contingency.
Milestone (month 9): measured pilot results and the first paid conversion.

dhaRti's risk is capped at ₹5L at a time: each tranche follows a met milestone. Can be funded directly or through NIDHI-PRAYAS / SISFS with dhaRti as host incubator.

QUALYXUS

Edge AI that traces every defect to its cause

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Demo
Live demo and code walkthrough available on request